

LC 53- maximum subarray

Approach:

- Kadane Algorithm finds the maximum sum contiguous subarray.
- Initialize count variable(say 'res') and maxsum=nums[0]
- Compute res by adding each array element of the array
- Update maxsum as max(maxsum,res)
- If res gets negative(<0), restart res=0
- Return maxsum

LC 75- sort colors

Approach:

- This follows bubble sort
- swapping element by creating a temporary space but that is not the intended solution.
- Or
- Instead creating a boundary for 0s,1s,2s is preferred as only 0,1,2 are the distinct elements in the array.
- low=0 for 0s, mid=0 for current index, high=n-1 for 2s
- traverse while mid<=high, if nums[mid]==0, swap with nums[low] and increment low and mid
- if nums[mid]==1, increment mid
- if nums[mid]==2, swap with nums[high], decrease high
- return nums

LC 121 Best time to buy stock and sell

Approach:

- This also follows Kadan's Algorithm, but instead of maximum subarray, we track the minimum difference
- Initialize minprice=nums[0] and profit=0
- Update minprice from min(minprice,prices[i]) and profit by max(profit,prices[i]-minprice)
- Return profit